Weil Conjectures

Chapter 17: The Deligne–Laumon theorem for curves

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17 The Deligne–Laumon theorem for curves

Let F be a pure lisse Weil sheaf of weight w on a smooth curve C. Why should $H^1_c(C, F)$ have weights at most $w + 1$? For the constant sheaf on a smooth projective curve this is Weil's theorem, proved with the Jacobian, and for an arbitrary F there is no such object to work with. Yet the Deligne–Laumon theorem comes down to exactly this bound. The theorem says that $H^r_c(C, F)$ has weights at most $w + r$ (Theorem 17.15), and only $r = 1$ requires work. Indeed, on an affine curve H^0_c vanishes, while by Lemma 15.7* H^2_c is a twist of the coinvariants of geometric monodromy, pure of weight $w + 2$. For $r = 1$ the bound can be strict: on $\mathbf{G}_m$ the group H^1_c comes from the boundary and has weight 0 (see the example of Chapter 13*).

For a Gauss sum the answer takes two steps. The transform $\hat{\chi}(y) = \overline{\chi}(y)g$ has the same absolute value for all $y \neq 0$, and Plancherel finds the common value $\sqrt{q}$. On an open of the affine line, Chapter 16* makes the same two steps available for sheaves. More precisely, let M be irreducible on a dense open $j : U_0 \hookrightarrow \mathbf{A}^1$, with no geometric constituent of the form $\mathcal{L}_{\psi(ax)}$, and let M' be the lisse sheaf with $\mathrm{FT}_\psi((j_*M)[1]) \simeq (j'_*M')[1]$ (Proposition 16.17*). Then

$$H^1_c(\mathbf{A}^1, j_*M) \simeq (j'_*M')_{0'}$$

(Corollary 17.6). Since a stalk of a middle extension has weights at most the weight of the lisse sheaf (Proposition 15.5*), it suffices to show that M' is pure of weight $w + 1$. Here the second step is the easy one, because once M' is pure, the sums of squared traces of Chapter 15* give its weight $w + 1$ (Theorem 17.5).

The first step, purity of M', is the deep one. The only purity theorem we have is Theorem 14.26*, for real sheaves. But the transform of a real sheaf is not real, since ψ takes nonreal values. For example, over $\mathbf{F}_3$ let M be the Kummer sheaf of the quadratic character χ on $\mathbf{G}_m$, which is real. By the computation of Chapter 16* and the signs in (16.3), the trace of M' at $y \in \mathbf{F}_3^\times$ is $-\chi(y)g$.

Here $g = \sum_x \chi(x)\psi(x) = \omega - \omega^2 = 1 + 2\omega$ is a square root of -3, which is not real. In this case, admittedly, a twist by $\mathcal{B}_{1/g}$ makes M' real, but in general no twist is known to do this.

The plan is to realize M', up to a twist, as a quotient of a real sheaf on the dual line, to which that theorem applies. To build it we add a conjugate. First we twist M to weight zero, $M^0 = M \otimes \mathcal{B}_b$. Then we transform $j_!M^0$ with the kernel $\mathcal{K}_\psi$ and $j_!(M^0)^\vee$ with the kernel $\mathcal{K}_{\psi^{-1}}$. At each point the traces of the two transforms are complex conjugate, so their sum is a real sheaf H on the dual line. Since the twist $M' \otimes \mathcal{B}_b$ is a quotient of H (Proposition 17.4), purity of real sheaves then makes M' pure. We use $j_!$ rather than j_* because its transform is a sheaf, and by Lemma 17.1 this transform differs from the transform of j_*M by a lisse sheaf of weights at most w.

Passing from one irreducible M to every pure sheaf needs care, because j_* is not exact in general. For example, the extension $0 \to \overline{\mathbf{Q}}_\ell(1) \to F \to \overline{\mathbf{Q}}_\ell \to 0$ on $\mathbf{G}_m$ given by the Kummer class of the coordinate is lisse, but $j_*F \to j_*\overline{\mathbf{Q}}_\ell$ is not surjective at 0, since the monodromy of F around 0 is a nontrivial unipotent matrix. For sheaves of one weight w, however, exactness holds by a weight gap (Proposition 17.9): by local duality at the boundary points (Lemma 17.7), the connecting map $j_*F_2 \to R^1j_*F_1$ goes from weights at most w to weights at least $w+2$, and so it vanishes.

17.1 Purity of the lisse part

Throughout this section and the next, $j : U_0 \hookrightarrow \mathbf{A}^1_{k_0}$ is a dense open and M is a nonzero irreducible lisse Weil sheaf on U_0, pure of weight w. We assume that no geometric irreducible constituent of M is the restriction of $\mathcal{L}_{\psi(ax)}$ for some $a \in \bar{k}_0$, and we let M' be the lisse Weil sheaf on a dense open $j' : U_0' \hookrightarrow \mathbf{A}^{1\vee}_{k_0}$ with $\mathrm{FT}_\psi((j_*M)[1]) \simeq (j'_*M')[1]$ (Proposition 16.17*).

We build the real sheaf from extensions by zero, because for them we can check reality on the trace function. Indeed, at a point of U_0 the eigenvalues of M^0 and of its dual have absolute value one, and hence they are complex conjugate. At a boundary point, however, the stalk of j_*M^0 is the space of inertia invariants, and the stalk of $j_*(M^0)^\vee$ is the space of invariants of the dual representation, but nothing we know at this stage makes their Frobenius eigenvalues conjugate. With $j_!$, by contrast, both stalks vanish there, and the trace function of the sum of the two transforms is clearly real.

The price is that $j_!M$ is not a simple perverse sheaf, so its transform G is not j'_*M' but an extension of j'_*M' by the transform B' of the boundary. This B' is a lisse sheaf of weights at most w (Lemma 17.1), and in the proof of Proposition 17.4 we use only that G is a sheaf and that it surjects onto j'_*M'.

Lemma 17.1 (Fourier transform of the boundary sequence)**.** *Put $B = j_*M/j_!M$. There are sheaves G and B' on the dual line such that $\mathrm{FT}_\psi(j_!M) = G$, $\mathrm{FT}_\psi(B) = B'[1]$, and the sequence*

$$0 \longrightarrow B' \longrightarrow G \longrightarrow j'_*M' \longrightarrow 0$$

is exact. The sheaf B' is lisse on the entire dual line and has weights at most w.

Proof. The three assertions are of different kinds. That the transform of $j_!M$ is a single sheaf is a vanishing statement for the cohomology of the twisted sheaves $M \otimes \mathcal{L}_{\psi(yx)}$ on U, whereas the exact sequence comes from transforming the boundary sequence $0 \to j_!M \to j_*M \to B \to 0$ term by term. Finally, the bound on B' comes from the stalks of B, which are known from Proposition 15.5*.

For G, the stalk formula of Proposition 16.5* says that the stalk of $\mathrm{FT}_\psi(j_!M)$ at y is $R\Gamma_c(U, M \otimes \mathcal{L}_{\psi(yx)})[1]$, so we must show that only H^1_c is nonzero. We apply Lemma 15.7* to the lisse sheaf

$M \otimes \mathcal{L}_{\psi(yx)}$ on the affine curve U, with j the identity. It shows that the group in degree zero vanishes and that the group in degree two is a twist of the coinvariants of geometric monodromy on $M \otimes \mathcal{L}_{\psi(yx)}$. But a nonzero coinvariant would be a geometric quotient of M isomorphic to $\mathcal{L}_{\psi(-yx)}$, and our assumption on M excludes such a constituent, so the group in degree two vanishes as well. The shift then places the remaining group H^1_c in degree zero, and the transform is a sheaf G.

The transform of B is computed point by point. After geometric base change, the punctual sheaf B is a finite sum of skyscrapers with coefficient modules, and the transform of each skyscraper is a linear character sheaf tensored with that module and shifted by one. Since a linear character sheaf is lisse on the whole dual line, $\mathrm{FT}_\psi(B) = B'[1]$ with B' lisse.

Applying the Fourier transform to the triangle obtained from $0 \to j_!M \to j_*M \to B \to 0$, and using Proposition 16.17* for the middle term, we obtain a triangle $G \to j'_*M' \to B'[1] \to$. Here G and j'_*M' are sheaves in degree zero, and $B'[1]$ is a sheaf in degree -1. Hence the long exact sequence of cohomology sheaves has only three nonzero terms, and it is the stated exact sequence.

The weights of B' come from those of B, because a character sheaf has weight zero, and the weights of B are at most w by Proposition 15.5*. Only at boundary points that are not rational does this need an argument. So let s be a boundary point of degree d and y a rational point of the dual line. The geometric points over s contribute d summands to $B'_{\bar{y}}$, which Frob_y permutes cyclically, and its dth power acts on one summand as Frob_s on $B_{\bar{s}}$ times a root of unity coming from the character sheaf. Every eigenvalue β of Frob_y on these summands therefore satisfies $\beta^d = \zeta\alpha$, with ζ a root of unity and α an eigenvalue of Frob_s. Since $|\iota\alpha| \le q^{dw/2}$, we get $|\iota\beta| \le q^{w/2}$. At a point y of higher degree we replace q by the cardinality of $\kappa(y)$.

For a filtration of B' with pure quotients, choose for each boundary point, over its residue field, a Frobenius-stable full flag of the coefficient module, which exists because $\overline{\mathbf{Q}}_\ell$ is algebraically closed. Each successive quotient is then a line, on which Frobenius acts by one of its eigenvalues. Tensoring with the linear character sheaves and summing over the Frobenius orbit, we obtain a finite Weil filtration of B' whose quotients are pure of the weights of the coefficient eigenvalues, and these weights are at most w. ∎

Example 17.2 (A Kummer sheaf has no boundary term)**.** Let M be the Kummer sheaf of a nontrivial character χ on $U_0 = \mathbf{G}_m$. It is irreducible and pure of weight 0. Since it is tame at infinity, it has no geometric constituent $\mathcal{L}_{\psi(ax)}$, and so it satisfies the hypotheses of this section. Inertia at the origin acts on M through χ, nontrivially, and hence $(j_*M)_{\bar{0}} = 0$ by Lemma 15.2*. This means that $j_!M = j_*M$ and $B = 0$, and the lemma gives $G \simeq j'_*M'$. On $\mathbf{F}_q$-points this is visible: by (16.3) the trace of G at $y \neq 0$ is $-\sum_x \chi(x)\psi(xy) = -\overline{\chi}(y)g$, with $g = \sum_x \chi(x)\psi(x)$, and for the quadratic character this is the trace of M' computed at the start of Chapter 17. At $y = 0$, on the other hand, the trace of G is $-\sum_x \chi(x) = 0$, in agreement with $H^1_c(\mathbf{G}_m, M) = 0$.

The argument of Proposition 15.9* also applies to twists by the Fourier kernel, and we need this in the proof of Proposition 17.4.

Lemma 17.3 (Twisting and adding the dual)**.** *Let $\mathcal{A}$ be lisse and pure of weight zero, and let $\mathcal{L}$ be lisse of rank one with local eigenvalues of finite order. Then*

$$(\mathcal{A} \otimes \mathcal{L}) \oplus (\mathcal{A}^\vee \otimes \mathcal{L}^{-1})$$

is real and pure of weight zero, and its extension by zero along an open immersion is real as well.

Proof. As in the proof of Proposition 15.9*, the local eigenvalues of $\mathcal{A} \otimes \mathcal{L}$ have absolute value one after applying ι, and those of the second summand are their inverses, hence their complex conjugates. The extension by zero is real as well, because its stalks outside the open are zero, and so its local characteristic polynomials there are equal to 1. ∎

In the next proof we use Lemma 17.1 twice. Applying its vanishing argument to M^0 and to its dual, we see that the transform of the real sheaf is a single sheaf. Then, using the surjection $G \to j'_* M'$ of the lemma, we exhibit a twist of M' as a quotient of that real sheaf.

Proposition 17.4 (Purity of the lisse part of the Fourier transform)**.** *The sheaf M' is pure of some real weight v on the whole open where it is lisse.*

Proof. We twist to weight zero and add the dual, as in Proposition 15.9*, before taking the transform, so that a twist of M' becomes a quotient of a real sheaf, to which Theorem 14.26* applies.

Choose b as in Proposition 15.9*, so that $M^0 = M \otimes \mathcal{B}_b$ is pure of weight zero.

On the product of the line and the dual line, consider the sheaf

$$F = (p_1^* j_! M^0 \otimes \mathcal{K}_\psi) \ \oplus \ (p_1^* j_! (M^0)^\vee \otimes \mathcal{K}_{\psi^{-1}}).$$

At each closed point, the Frobenius eigenvalues of the second summand are the complex conjugates of those of the first, because the eigenvalues of M^0 have absolute value one and those of the character sheaves are roots of unity, while at the boundary both zero extensions have zero stalks. Hence F is real, as in Lemma 17.3. Moreover, since duality replaces a by $-a$, the dual $(M^0)^\vee$ also has no geometric constituent of the form $\mathcal{L}_{\psi(ax)}$, so the argument of Lemma 17.1 shows that the direct image with compact support of F along p_2 is concentrated in degree one. We set $H = R^1 p_{2!} F$.

Let y be a point with values in a finite field. Since $R p_{2!} F = H[-1]$, the trace formula for $R p_{2!}$ (Corollary 12.26*) gives $-t_H(y) = \sum_x t_F(x, y)$. For each x the traces of the two summands of F at (x, y) are complex conjugate, so $t_H(y)$ is real. Over the finite extensions of $\kappa(y)$ the same argument shows that the trace of every power of the local Frobenius is real. These traces are the power sums of the local eigenvalues, and the Newton identities express the coefficients of the characteristic polynomial through the power sums with rational coefficients. Hence the local characteristic polynomials of H have real coefficients, that is, H is real.

The first summand of H is $R^1 p_{2!}(p_1^* j_! M \otimes \mathcal{K}_\psi) \otimes \mathcal{B}_b$ by the projection formula for the constant twist, that is, $G \otimes \mathcal{B}_b$ with $G = \mathrm{FT}_\psi(j_! M)$ as in Lemma 17.1. Tensoring the surjection $G \to j'_* M'$ of that lemma with $\mathcal{B}_b$, we see that $M' \otimes \mathcal{B}_b$ is a quotient of a direct summand of H. To apply the purity theorem we restrict to a dense open on which both H and M' are lisse. There $M' \otimes \mathcal{B}_b$ is a lisse quotient of the real lisse sheaf H, and it is irreducible by Lemma 16.14*, so by Theorem 14.26* it is pure of some real weight. Removing the twist by $\mathcal{B}_b$, we find that M' is pure of some weight v on that open. Since the open where M' is lisse is a smooth geometrically connected curve, the last assertion of Corollary 15.6* extends the purity to all of it. ∎

17.2 The weight of the Fourier transform

By Proposition 17.4, M' is pure of some weight v, and Plancherel now finds v. Purity had to come first, since the growth exponent ν of Definition 15.10* is the weight only for pure sheaves.

Theorem 17.5 (The weight of the Fourier transform)**.** *The sheaf M' is pure of weight exactly $w + 1$.*

Proof. Let v be the weight of M' given by Proposition 17.4. Shifts do not change the squared absolute values of traces, and, if a nonzero lisse sheaf on a dense open of the line is pure of weight u, then its direct image j_* has $\nu = u$ (Corollary 15.15*). Applying this to M' and to M, and using the Plancherel formula (Corollary 16.8*), we get

$$v = \nu(j'_*M') = \nu(\mathrm{FT}_\psi((j_*M)[1])) = \nu((j_*M)[1]) + 1 = w + 1.$$

The first equality holds because M' is pure of weight v, the second is the defining isomorphism of M', the third is Plancherel, and the last holds because M is pure of weight w. ∎

In the example of the chapter opening, M is the Kummer sheaf of the quadratic character over $\mathbf{F}_3$, of weight 0, and the trace of M' at $y \in \mathbf{F}_3^\times$ is $-\chi(y)g$ with $g^2 = -3$. Its absolute value is $\sqrt{3} = 3^{1/2}$, which is weight $1 = w + 1$, as the theorem predicts.

The weight of M' bounds H^1_c, because the stalk of the transform at the origin of the dual line is the compactly supported cohomology of j_*M.

Corollary 17.6 (H^1_c as a stalk of the Fourier transform)**.** *We have $H^2_c(\mathbf{A}^1, j_*M) = 0$, and there is a Frobenius-equivariant isomorphism $H^1_c(\mathbf{A}^1, j_*M) \simeq (j'_*M')_{0'}$ with the stalk at the origin $0'$ of the dual line. In particular $H^1_c(\mathbf{A}^1, j_*M)$ has weights at most $w + 1$.*

Proof. We compute the stalk at $0'$ of $\mathrm{FT}_\psi((j_*M)[1])$ in two ways: by the stalk formula, which gives the cohomology of j_*M, and through the isomorphism with $(j'_*M')[1]$, which gives a stalk of a middle extension.

At the rational origin $0'$ of the dual line, the kernel $\mathcal{K}_\psi$ restricts to the pullback of $\mathcal{L}_\psi$ along the constant map to 0, which is canonically trivial (on traces this is $\psi(0) = 1$). The stalk formula of Proposition 16.5*, which is compatible with Frobenius, therefore gives

$$\big(\mathrm{FT}_\psi((j_*M)[1])\big)_{0'} = R\Gamma_c(\mathbf{A}^1, j_*M)[2].$$

We compare this with the stalk of $(j'_*M')[1]$, which is concentrated in degree -1. In degree 0 we therefore see that $H^2_c(\mathbf{A}^1, j_*M) = 0$, and in degree -1 we obtain

$$H^1_c(\mathbf{A}^1, j_*M) \simeq (j'_*M')_{0'} = (M'_{\bar\eta})^{I_{0'}},$$

where $I_{0'}$ is the inertia group at the origin $0'$ of the dual line acting on the generic fibre of M', and the last equality is the formula of Lemma 15.2* for the stalk of a middle extension. By Proposition 15.5* the weights of this stalk are at most the weight of M', which is $w + 1$ by Theorem 17.5. ∎

17.3 The affine line

To pass from irreducible sheaves to arbitrary ones we show that j_* is exact on sequences of pure sheaves of one weight. The reason is a gap in weights. Let $0 \to F_1 \to F \to F_2 \to 0$ be an exact sequence of lisse sheaves on a dense open U of a curve C. Applying j_* gives an exact sequence $0 \to j_*F_1 \to j_*F \to j_*F_2 \to R^1j_*F_1$, and $R^1j_*F_1$ is supported on $C \setminus U$. Now suppose that F_1 and F_2 are pure of weight w. Then, by Proposition 15.5*, the source of the last map has weights at most w at a point of $C \setminus U$. On the other hand, by local duality, which we prove first, the stalk of $R^1j_*F_1$ at such a point is the dual of the invariants of $F_1^\vee$, twisted by (-1), and this

space has weights at least $w+2$. Since a Frobenius-equivariant map between spaces with disjoint weights is zero, j_* is exact on such sequences (Proposition 17.9).

The hypothesis of equal weights is needed. In the Kummer extension $0 \to \overline{\mathbf{Q}}_\ell(1) \to F \to \overline{\mathbf{Q}}_\ell \to 0$ of the chapter opening, $F_1 = \overline{\mathbf{Q}}_\ell(1)$ has weight -2 and $F_2 = \overline{\mathbf{Q}}_\ell$ has weight 0, so at the origin the stalks of $j_*\overline{\mathbf{Q}}_\ell$ and of $R^1 j_*\overline{\mathbf{Q}}_\ell(1)$ both have weight 0, and the connecting map is indeed nonzero.

Exactness lets us extend Corollary 17.6 from one irreducible sheaf with no geometric constituent of the form $\mathcal{L}_{\psi(ax)}$ to a pure sheaf unramified at infinity (Theorem 17.10) by induction along a composition series. Each factor of the series is either geometrically constant, and then $H^1_c = 0$, or it satisfies the hypotheses of the corollary.

Lemma 17.7 (Local duality at a point of $C \setminus U$)**.** *Let $j_0 : U_0 \hookrightarrow C_0$ be a dense open in a smooth curve over k_0, and let F be a lisse Weil $\overline{\mathbf{Q}}_\ell$-sheaf on U_0. For a geometric point $\bar{s}$ over a closed point $s \in C_0 \setminus U_0$, there is a canonical perfect pairing, equivariant for the local geometric Frobenius at s,*

$$(j_*F^\vee)_{\bar{s}} \otimes (R^1 j_* F)_{\bar{s}} \longrightarrow \overline{\mathbf{Q}}_\ell(-1).$$

Proof. The two stalks are cohomology sheaves of the complexes $i^*Rj_*F^\vee$ (in degree zero) and i^*Rj_*F (in degree one) on the boundary. We compute the Verdier dual of the second complex by duality on the curve, and we find the first one, shifted and twisted.

Let $i : Z \hookrightarrow C$ be the inclusion of the whole finite complement, and put $K_F = i^*Rj_*F$. For a lisse sheaf G on U, the adjunction $j^* \dashv Rj_*$ shows that $i^!Rj_*G = 0$, and applying $i^!$ to the localization triangle for Rj_*G we obtain $i^!j_!G \simeq i^*Rj_*G[-1]$. Since duality exchanges Rj_* with $j_!$ and i^* with $i^!$, this formula lets us write the result again in the form i^*Rj_*. Indeed, by Verdier duality on the smooth curve, $\mathbb{D}_U F = F^\vee(1)[2]$ for the lisse sheaf F, and we find

$$\begin{aligned}\mathbb{D}_Z K_F &\simeq i^!\mathbb{D}_C Rj_*F \simeq i^!j_!\mathbb{D}_U F \\ &\simeq i^*Rj_*(F^\vee(1)[2])[-1] = K_{F^\vee}(1)[1].\end{aligned}$$

On the finite geometric scheme Z, Verdier duality is the exact duality of coefficient vector spaces, so taking degree -1 at $\bar{s}$ we get $(R^1 j_*F)^\vee_{\bar{s}} = (j_*F^\vee)_{\bar{s}}(1)$, which is the pairing. It is Frobenius-equivariant because the adjunction, localization and duality maps are natural in F and respect its Weil structure. ∎

For F pure lisse of weight w, the pairing shows that $(R^1 j_*F)_{\bar{s}}$ has weights at least $w+2$. Indeed, it puts this stalk in duality with a stalk of $j_*F^\vee$, which has weights at most $-w$, and the pairing takes values in $\overline{\mathbf{Q}}_\ell(-1)$, which has weight 2. The next proposition uses this gap between the source and the target of the connecting map.

Example 17.8 (Local duality at the origin of the line)**.** Let $j : \mathbf{G}_m \hookrightarrow \mathbf{A}^1$ over k_0, let $F = \overline{\mathbf{Q}}_\ell$, which is pure of weight 0, and let $s = 0$. Since inertia at the origin acts trivially on the constant sheaf, $(j_*\overline{\mathbf{Q}}_\ell)_{\bar{0}} = \overline{\mathbf{Q}}_\ell$ by Lemma 15.2*, with Frobenius acting by 1, of weight 0. On the other side, the Kummer covers $t \mapsto t^{\ell^n}$ of a punctured neighbourhood of the origin give $(R^1 j_*\overline{\mathbf{Q}}_\ell)_{\bar{0}} = \overline{\mathbf{Q}}_\ell(-1)$ (Chapter 6*), on which geometric Frobenius acts by q, of weight 2. The pairing of Lemma 17.7 is the multiplication $\overline{\mathbf{Q}}_\ell \otimes \overline{\mathbf{Q}}_\ell(-1) \to \overline{\mathbf{Q}}_\ell(-1)$. Thus the two stalks have weights 0 and 2, and the gap of two comes entirely from the Tate twist. Indeed, with coefficients $\overline{\mathbf{Q}}_\ell(1)$, H^1 of a punctured disc is spanned by the Kummer class of the coordinate, which Frobenius fixes, so with coefficients $\overline{\mathbf{Q}}_\ell = \overline{\mathbf{Q}}_\ell(1) \otimes \overline{\mathbf{Q}}_\ell(-1)$ Frobenius acts by q.

Proposition 17.9 (Exactness at a fixed pure weight)**.** *Let $j_0 : U_0 \hookrightarrow C_0$ be as in Lemma 17.7. If $0 \to F_1 \to F \to F_2 \to 0$ is a short exact sequence of lisse Weil sheaves on U_0, with F_1 and F_2 pure of the same weight w, then the sequence*

$$0 \longrightarrow j_*F_1 \longrightarrow j_*F \longrightarrow j_*F_2 \longrightarrow 0$$

is exact.

Proof. Applying j_* to the short exact sequence gives the long exact sequence

$$0 \to j_*F_1 \to j_*F \to j_*F_2 \xrightarrow{\delta} R^1 j_*F_1,$$

so it suffices to show that the connecting map δ vanishes. It is Frobenius-equivariant, and we show that its source and target have disjoint weights. Since $R^1 j_*F_1$ is zero on U, we look at a point of $C \setminus U$. There the stalk of the source has weights at most w by Proposition 15.5*. For the target, note that the dual $F_1^\vee$ is pure of weight $-w$, so the same result shows that the stalk of $j_*F_1^\vee$ has weights at most $-w$. By Lemma 17.7 the stalk of the target is the dual of this stalk twisted by (-1), and hence the target has weights at least $w+2$. Now the image of δ is a quotient of the source and a submodule of the target, so its eigenvalues would have weights both at most w and at least $w+2$. Hence δ vanishes at each point of $C \setminus U$. As $R^1 j_*F_1$ is zero on U, it follows that $\delta = 0$. ∎

All factors of a composition series of a pure lisse sheaf are pure of the same weight, so the proposition applies to every step of such a series. In this way we pass from the irreducible sheaves of Corollary 17.6 to arbitrary ones.

Theorem 17.10 (The weight bound on the affine line)**.** *Let $j : U_0 \hookrightarrow \mathbf{A}^1_{k_0}$ be a dense open, and let F be a lisse sheaf on U_0, pure of weight w and unramified at infinity. Then $H^1_c(\mathbf{A}^1, j_*F)$ has weights at most $w+1$.*

Proof. We prove the bound for the irreducible factors of F and pass to F by induction. The irreducible factors are of two kinds: geometrically constant ones, for which H^1_c vanishes, and the others, to which Corollary 17.6 applies.

Take a finite composition series of the finite-dimensional Weil representation of F. Each irreducible lisse factor is pure of weight w, since its local eigenvalues are eigenvalues of a subquotient. It is again unramified at infinity, since inertia at infinity acts trivially on every subquotient of a representation on which it acts trivially. An irreducible factor that is geometrically constant is a sheaf $\mathcal{B}_b$ of Proposition 13.7*, because the geometric action is trivial and an invertible linear operator over $\overline{\mathbf{Q}}_\ell$ has an eigenline. Its middle extension is then constant on the affine line, and so its H^1_c vanishes by the trivial case of Lemma 16.4*. Every other factor F_i is irreducible, unramified at infinity and not geometrically constant, so by Lemma 16.18* it has no geometric constituent of the form $\mathcal{L}_{\psi(ax)}$, and we may apply Corollary 17.6 with $M = F_i$.

The induction needs j_* to turn the composition series into a filtration of j_*F whose quotients are the j_*F_i. Since all factors are pure of the same weight w, Proposition 17.9 shows that the short exact sequences of the series remain exact after applying j_*. So consider a step $0 \to F' \to F'' \to F_i \to 0$ of the series, where we may assume that the bound holds for F'. In the resulting long exact sequence of compactly supported cohomology, the image of $H^1_c(\mathbf{A}^1, j_*F')$ in $H^1_c(\mathbf{A}^1, j_*F'')$ is a quotient of the former, and the cokernel embeds in $H^1_c(\mathbf{A}^1, j_*F_i)$. Since the bound $w+1$ is preserved under subquotients and extensions, we conclude by induction that it holds for F. ∎

The condition at infinity enters only through Lemma 16.18*, to verify the hypothesis of Proposition 16.17*: it excludes the constituents $\mathcal{L}_{\psi(ax)}$ with $a \neq 0$, whose transforms are skyscrapers rather than middle extensions. In the last section we remove this hypothesis, together with the restriction to opens of the line.

17.4 Arbitrary curves

Theorem 17.10 assumes that the curve is an open of the affine line and that the sheaf is unramified at infinity. We remove the second hypothesis by moving a point where the sheaf is lisse to infinity, and the first by taking the direct image along a finite map to the line.

For an upper bound it is enough to replace H^1_c by a group that surjects onto it, and each move does this. For the first move, let s be a point where the sheaf is lisse. Since a point has no H^1_c, deleting s gives a surjection $H^1_c(W \setminus \{s\}, F) \to H^1_c(W, F)$, and after a finite extension of the constant field and an automorphism of $\mathbf{P}^1$ sending s to infinity, the sheaf is unramified there. For the second, a finite map $\pi : C \to \mathbf{A}^1$ does not change compactly supported cohomology at all, since $R\Gamma_c(\mathbf{A}^1, \pi_* j_! F) = R\Gamma_c(C, F)$. However, $\pi_* j_! F$ is lisse and pure only off the branch locus, so we remove the bad points by one more localization. Such a map π exists by Riemann–Roch, because a function with a single pole of order prime to p defines a finite generically étale map (Lemma 17.13). Curves that are not affine, and the degrees $r \neq 1$, are treated afterwards in Theorem 17.15.

Lemma 17.11 (Weight bounds and finite constant extensions)**.** *Let X_0 be a separated scheme of finite type over k_0, let F be a constructible Weil sheaf on X_0, let $k_m \supset k_0$ be the extension of degree m, and let $u \in \mathbf{R}$. The group $H^r_c(X, F)$ is the same for X_0 and for $X_0 \otimes_{k_0} k_m$, and the geometric Frobenius of k_m acts on it as the mth power of that of k_0. Every eigenvalue α of the Frobenius of k_0 on $H^r_c(X, F)$ satisfies $|\iota\alpha| \leq q^{u/2}$ if and only if every eigenvalue β of the Frobenius of k_m satisfies $|\iota\beta| \leq q^{mu/2}$.*

Proof. An extension of degree m of the constant field does not change the geometric scheme X, the sheaf on it, or its cohomology, and it replaces the Frobenius of k_0 by its mth power. So the eigenvalues β are the α^m, and $|\iota\alpha^m| \leq q^{mu/2}$ holds if and only if $|\iota\alpha| \leq q^{u/2}$. ∎

To move a point to infinity we need a rational point, and hence a finite extension of the constant field. By the lemma this does not affect a weight bound.

Lemma 17.12 (Moving a lisse point to infinity)**.** *Let W_0 be a dense open of $\mathbf{P}^1_{k_0}$, and let F be a lisse Weil sheaf on W_0, pure of weight w. There are a finite extension k_m of k_0, a dense open V_0 of $\mathbf{A}^1_{k_m}$, a lisse Weil sheaf G on V_0, pure of weight w and unramified at infinity, and a surjection*

$$H^1_c(V, G) \twoheadrightarrow H^1_c(W, F)$$

that commutes with the geometric Frobenius of k_m.

Proof. We delete a rational point where F is lisse, which replaces H^1_c by a group surjecting onto it, and move that point to infinity by an automorphism of $\mathbf{P}^1$. First choose a closed point of W_0 and a finite extension k_m of the constant field over which it has a rational point s. The restriction of F to $W_0 \otimes_{k_0} k_m$ is again pure of weight w, because a residue extension raises an eigenvalue and the residue cardinality to the same power. Next, some element of $\mathrm{PGL}_2(k_m)$ sends s to infinity, and we transport $F|_{W \setminus \{s\}}$ by this automorphism. As F is lisse at s, its local inertia there is trivial, so the transported sheaf G extends to a lisse sheaf across the new infinity. It is defined

on the image V_0 of $W_0 \setminus \{s\}$, a dense open of the affine line, and $H^1_c(V, G) = H^1_c(W \setminus \{s\}, F)$ because the automorphism identifies the two pairs. The surjection follows from the localization sequence

$$H^1_c(W \setminus \{s\}, F) \to H^1_c(W, F) \to H^1_c(\{s\}, F_{\bar{s}}) = 0,$$

in which the last group vanishes because a point has no cohomology in positive degrees. Since every construction is defined over k_m, the map commutes with the Frobenius of k_m. ∎

By the lemma, for any open of the projective line and any pure lisse sheaf on it, we may bound H^1_c with Theorem 17.10. To reach an arbitrary affine curve we map it to the line, and we have to choose the map so that the direct image is again pure and lisse on a dense open.

Lemma 17.13 (A finite generically étale map to the line)**.** *Let C_0 be a smooth geometrically connected affine curve. There are a smooth curve C_0' containing C_0 with finite complement and a finite generically étale map $\pi : C_0' \to \mathbf{A}^1_{k_0}$. If F is a pure lisse sheaf of weight w on C_0, then the sheaf $K = \pi_* j_! F$, where $j : C_0 \hookrightarrow C_0'$, is lisse and pure of weight w outside the branch locus and the image of $C_0' \setminus C_0$. Moreover, $R\Gamma_c(\mathbf{A}^1, K) = R\Gamma_c(C, F)$ canonically.*

Proof. The map is given by a function with a single pole, at a point at infinity, and we choose the order of this pole prime to p so that the map is generically étale. We find the function by Riemann–Roch, we show by a local computation at the pole that its differential is nonzero, and we check the assertions about K stalk by stalk.

Let $\overline{C}_0$ be the smooth projective completion, choose a closed point s of $\overline{C}_0$ outside C_0, and choose m prime to p with $(m-1)\deg(s) > 2g(\overline{C}_0) - 2$. Both divisors ms and $(m-1)s$ then have degree $> 2g-2$, so Riemann–Roch (Theorem C.3*) gives $h^0(\mathcal{O}(ns)) = n\deg(s) + 1 - g$ for $n = m, m-1$, and hence

$$h^0(\mathcal{O}(ms)) - h^0(\mathcal{O}((m-1)s)) = \deg(s) > 0.$$

Therefore there is a function f whose only pole is at s, of exact order m. The sections $1, f$ define a nonconstant map $\overline{C}_0 \to \mathbf{P}^1$ whose inverse image of infinity is $\{s\}$, and this map is proper with finite fibres, hence finite. To see that it is generically étale, choose a uniformizer t at s and write $f = t^{-m}u$ with u a unit. Since $\kappa(s)/k_0$ is separable and the curve is smooth, dt generates the module of differentials near s. Writing $du = u'\,dt$, we get

$$df = t^{-m-1}(-mu + tu')\,dt \neq 0,$$

since $-mu + tu'$ is congruent to the unit $-mu$ modulo t, and mu is a unit because $p \nmid m$. The differential of this finite map between smooth curves is therefore invertible at the generic point, and the map is generically étale. We set $C_0' = \overline{C}_0 \setminus \{s\}$ and let π be the restriction of the map to C_0'.

Over the open in the statement, π is finite étale and its inverse image lies in C_0, where $j_! F = F$. Moreover, the direct image of a lisse sheaf under a finite étale map is lisse, because étale locally the map is a disjoint union of copies of the base. Hence K is lisse there. For purity, note that at a closed point y of this open, the stalk of K is a sum over the geometric points of the fibre, which Frobenius permutes. Consider such a cycle of length e, corresponding to a point over y with residue extension of degree e. On it the eth power of the operator is the local Frobenius upstairs, so an eigenvalue β satisfies $|\iota\beta|^e = q_y^{ew/2}$, and the weight is still w. Since $R\pi_! = \pi_*$

for the finite map π, the identification of cohomology follows by composing direct images with compact support. It commutes with Frobenius because the construction is defined over k_0. ■

For example, take for C_0 the affine curve $y^2 = x^3 - x$ over $\mathbf{F}_3$, the complement of the point at infinity of the projective curve. There $g = 1$, the point $s = \infty$ is rational, and $m = 2$ satisfies $(m-1)\deg(s) > 0$. Accordingly, the function x has a single pole, of order two, which is prime to 3, and the map is the double cover $(x, y) \mapsto x$ of the line, whose extension to the projective curve is branched over 0, ± 1 and ∞.

With both lemmas the bound on the line extends to every affine curve.

Theorem 17.14 (The weight bound on an affine curve)**.** *For a pure lisse Weil sheaf F of weight w on a smooth geometrically connected affine curve C_0, the group $H^1_c(C, F)$ has weights at most $w + 1$.*

Proof. We push the sheaf forward to the line and realize H^1_c as a quotient of a group bounded by Theorem 17.10. This uses two surjections: one removes the points where the direct image fails to be lisse and pure, and the other moves a lisse point to infinity.

Choose π as in Lemma 17.13 and put $K = \pi_* j_! F$. Let W_0 be the dense open on which K is lisse and pure. Since its complement is finite, localization gives

$$H^1_c(W, K|_W) \twoheadrightarrow H^1_c(\mathbf{A}^1, K) = H^1_c(C, F).$$

By Lemma 17.12, after a finite extension of the constant field there is a further surjection from $H^1_c(V, G)$ onto $H^1_c(W, K|_W)$, with G pure of weight w and unramified at infinity. Let $v : V \hookrightarrow \mathbf{A}^1$ be the inclusion and $i : Z \hookrightarrow \mathbf{A}^1$ its finite complement. The localization sequence for $v_* G$ reads

$$H^0_c(Z, i^* v_* G) \longrightarrow H^1_c(V, G) \longrightarrow H^1_c(\mathbf{A}^1, v_* G) \longrightarrow 0.$$

The first group has weights at most w. Indeed, the stalks of $i^* v_* G$ at the closed points of Z_0 have weights at most w by Proposition 15.5*, and $H^0_c(Z, i^* v_* G)$ is the direct sum, over the closed points z of Z_0, of the stalks at the $e = [\kappa(z) : k_m]$ geometric points over z, where k_m is the constant field of V_0. The geometric Frobenius of k_m permutes these e stalks cyclically, and its eth power on each of them is conjugate to the local Frobenius F_z. An eigenvalue β on such a cycle therefore satisfies $\beta^e = \alpha$ for an eigenvalue α of F_z, so that $|\iota\beta|^e = |\iota\alpha| \leq q_z^{w/2} = (q^m)^{ew/2}$, and $|\iota\beta| \leq (q^m)^{w/2}$, which is weight at most w over k_m. This is the case of the finite map $Z_0 \to \operatorname{Spec} k_m$ in Lemma 18.1*, whose proof is the same computation. The last group has weights at most $w + 1$ by Theorem 17.10. As an extension of a subgroup of the last by a quotient of the first, the middle group therefore has weights at most $w + 1$. This bound passes to quotients under the two surjections, and by Lemma 17.11 it holds over the original constant field. ■

To reach a curve that is not affine, we pass to a dense affine open and view the sheaf on the whole curve as a middle extension.

Theorem 17.15 (The weight bound on a curve)**.** *Let $j : U_0 \hookrightarrow C_0$ be a dense open in a smooth curve over k_0, and let F be a pure lisse Weil sheaf of weight w on U_0. Then for every $r \in \mathbf{Z}$ the group $H^r_c(C, j_* F)$ has weights at most $w + r$. In particular, for $U_0 = C_0$, the group $H^r_c(C, F)$ has weights at most $w + r$.*

Proof. On a geometrically connected curve we compare j_*F with its restriction to a dense affine open, where the affine case applies, and we deduce the bounds from the localization sequence. The general case reduces to this one, because a curve that is not geometrically connected becomes a disjoint union of geometrically connected ones after a finite constant extension.

Suppose first that C_0 is geometrically connected, and choose a dense affine open $v : V_0 \hookrightarrow U_0$. Since F is lisse on U_0, the unit $F \to v_*F|_V$ is an isomorphism by the last assertion of Lemma 15.2*. Hence j_*F is also the middle extension of $F|_V$ from V_0 to C_0. On V, by Theorem 17.14 and Lemma 15.7*, $H^r_c(V, F)$ has weights at most $w + r$ for every r, and $H^0_c(V, F) = 0$. On the complement, write $Z_0 = C_0 \setminus V_0$ and i for its inclusion; then the restriction of j_*F to the finite set Z_0 has weights at most w by Proposition 15.5*, and its cohomology is concentrated in degree zero. These two pieces are joined by the localization sequence for $V \subset C$, which reads

$$0 = H^0_c(V, F) \to H^0_c(C, j_*F) \to H^0(Z, i^*j_*F) \to H^1_c(V, F) \to H^1_c(C, j_*F) \to 0,$$

followed by an isomorphism $H^2_c(V, F) \simeq H^2_c(C, j_*F)$, and all higher groups vanish. Over $\operatorname{Spec} k_0$ the bounds are conditions on eigenvalues (Lemmas 13.2* and Lemma 13.3*), and we read them off degree by degree. In degree zero, the group $H^0_c(C, j_*F)$ injects into $H^0(Z, i^*j_*F)$, which has weights at most w: Frobenius permutes the stalks at the geometric points over each closed point of Z_0 cyclically, with the local Frobenius as the power that returns to a stalk, as in the proof of Theorem 17.14. In degree one, $H^1_c(C, j_*F)$ is a quotient of $H^1_c(V, F)$, which has weights at most $w + 1$. In degree two the groups are isomorphic, and the bound $w + 2$ holds for V.

In general, the connected components of C are defined over a finite extension k_m of k_0, and over k_m each of them is a smooth geometrically connected curve. For example, $\mathbf{P}^1_{\mathbf{F}_{q^2}}$, viewed as a curve over $\mathbf{F}_q$, is connected, but C has two components, which Frobenius exchanges and which are defined over $\mathbf{F}_{q^2}$. As Frobenius permutes the components, only its mth power acts on the cohomology of each of them. On such a component, j_*F is the middle extension of the restriction of F to the dense open where the component meets U, and this restriction is pure of weight w. Moreover, the group $H^r_c(C, j_*F)$ is the direct sum of the groups of these components, compatibly with the mth power of Frobenius. Hence the bound holds over k_m by the first case, and by Lemma 17.11 it holds over k_0. ∎

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